

PSD: Parallel Finite Element Solver for Continuum Dynamics

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Summary

PSD (Parallel finite element Solver for continuum Dynamics) is an open-source finite element method (FEM) solver designed for high-performance computing simulations in continuum dynamics with a special focus on earthquake mechanics. It enables integrated fault-to-site seismic simulations by combining advanced material modeling, scalable parallelism, and purpose-built meshing-partitioning tools.

Built on FreeFEM ([Hecht, 2012](#)) for FEM discretization and PETSc ([Balay et al., 2019](#)) for scalable linear solvers, PSD integrates non-linear material modeling through its MFronT ([Helfer et al., 2015](#)) interface for realistic simulations. Its custom MPI I/O-based mesher-partitioner, top-ii-vol ([Badri et al., 2024](#)), enables efficient on-the-fly mesh generation and partitioning for earthquake geometries, removing sequential meshing bottlenecks. On the structural side, hybrid phase-field fracture mechanics ([Ambati et al., 2015](#)) for crack analysis is implemented. This spans the full simulation chain from earthquake source to structural assessment.

A key feature of PSD is its ability to perform fault-to-site earthquake simulations with billions of degrees of freedom, scaling efficiently on tens of thousands of MPI-processes, making it suitable for comprehensive seismic risk assessment.

Statement of Need

Seismic risk assessment requires tools that can simulate wave propagation across multiple scales, from faults (kilometers away) to local site response (meters), with sufficient accuracy. Commercial software often lacks the scalability needed for regional simulations, while open-source alternatives typically focus on specific parts of the seismic workflow.

Current computational challenges in earthquake simulation include: (1) handling billions of degrees-of-freedom to capture realistic fault-to-site scenarios ([Hori et al., 2018](#)), ([Cui et al., 2013](#)), (2) integrating complex non-linear material behaviors and damage assessment for solids and structures, and (3) efficiently generating and partitioning large meshes for irregular geological domains derived from digital elevation models. FEM tools like OpenSees ([McKenna, 2011](#)) excel in local-site response, while SPECfem3D ([Peter et al., 2011](#)) and SEM3D ([Touhami et al., 2022](#)) address wave propagation using spectral elements. An open-source platform covering the full fault-to-site workflow with HPC scalability remains highly desirable.

PSD tries to fill this gap by providing a unified framework that combines earthquake wave propagation simulation and structural mechanics assessment within a single scalable FEM solver. PSD's integration of advanced meshing-partitioning capabilities, sophisticated material modeling, and fracture mechanics positions it uniquely for comprehensive seismic risk assessment needs

41 requiring both regional-scale wave propagation and local site response, including structural
42 analysis.

43 Features and Architecture

44 PSD offers a range of physics modules designed for earthquake simulations, including linear
45 elasticity, elastodynamics, fracture mechanics, soildynamics¹, and elasto-plasticity. Its versatility
46 is further enhanced by the Mfront interface, which allows users to implement custom non-linear
47 material models, that can be seamlessly integrated into any of these modules expanding their
48 capabilities beyond its built-in constitutive laws. Additionally, comprehensive verification and
49 validation campaigns for all PSD modules, with results cross-compared against reference codes,
50 experimental data, and analytical benchmarks, ensuring transparency and reproducibility (see
51 [validation page](#)).

52 PSD adopts a layered architecture that separates mathematical formulation from computational
53 implementation while maintaining high performance through strategic integration. It follows a
54 code generation approach, users specify problem configurations through command-line options,
55 after which PSD_PreProcess generates optimized FreeFEM code tailored to the selected physics,
56 dimensionality, and boundary conditions. The generated code is coupled with PSD-specific
57 compiled components, including physics modules, MFront interfaces, MED-based pre- and
58 post-processing tools, distributed mesh generation and partitioning utilities, nonlinear soil
59 constitutive models, and eigensolver components for fracture mechanics. This complete
60 software stack defines the governing formulations, assembles the corresponding finite-element
61 problems, integrates specialized material models and boundary conditions, manages distributed
62 meshes, invokes the required numerical libraries, and controls the simulation workflow. This
63 design enables computational efficiency while preserving flexibility for diverse applications
64 across the available physics modules.

65 PSD adopts a layered architecture that separates mathematical formulation from computational
66 implementation while maintaining high performance through strategic integration. It follows a
67 code generation approach, users specify problem configurations through command-line options,
68 after which PSD_PreProcess generates optimized FreeFEM code tailored to the selected physics,
69 dimensionality, and boundary conditions. The generated FreeFEM code is not intended to run
70 only as a standalone script for an unmodified FreeFEM installation. Instead, it is coupled with
71 PSD-specific compiled plugins and software components, including: - MPI-I/O-based distributed
72 point-cloud-to-volume mesh generation and partitioning utilities for earthquake simulations; -
73 compiled interfaces to MFront for the assembly of nonlinear constitutive behaviours; - MED-based
74 mesh and post-processing interfaces, including data transfer between earthquake and structural
75 simulations; - dedicated constitutive models and special boundary conditions for soil mechanics;
76 - eigensolver components used in phase-field fracture-mechanics formulations.

77 Therefore, PSD should be understood as a complete solver stack: it defines the governing
78 formulations, assembles the corresponding finite-element problems, integrates specialised
79 material models and boundary conditions, manages distributed meshes, invokes the required
80 numerical libraries, and controls the simulation workflow. In this stack, FreeFEM provides
81 the general finite-element language and discretization kernel, while the solver functionality is
82 provided by the combination of generated code, PSD plugins, physics modules, and execution
83 tools. This design enables computational efficiency while preserving flexibility for diverse
84 applications across the available physics modules.

85 The parallel computing architecture in PSD employs domain decomposition strategies that
86 enable distributed memory parallelization which are optimized for large-scale FEM simulation

¹The *soildynamics* module builds upon the *elastodynamics* module by adding tools essential for earthquake modeling, such as paraxial boundary conditions, double-couple source mechanisms, point-cloud meshing-partitioning, etc.

(Dolean et al., 2015). PSD has demonstrated scalability up to 24,000 cores and capability for handling problems with over 5 billion unknowns for earthquakes.

Example Workflow

A representative application illustrates PSD's soildynamics module for 3D wave propagation in an elastic domain with paraxial absorbing boundaries (Modaressi & Benzenati, 1994) and double-couple point sources (Benz & Smith, 1987). This example illustrates one of PSD's specialized physics modules, among others. The aim here is to briefly illustrate PSD's key capabilities, including automated distributed mesh generation, advanced time integration, and sophisticated boundary condition handling.

Mathematical Presentation: PSD solves the elastodynamic wave equation using FEM discretization with Newmark- β time integration. For a domain $\Omega \subset \mathbb{R}^3$ and with paraxial absorbing boundaries $\partial\Omega_p \subset \partial\Omega$, the FEM weak form reads:

Find $\mathbf{u} \in \mathcal{U}$ such that $\forall t \in [0, t_{\max}], \forall \mathbf{v} \in \mathcal{V}$:

$$\begin{aligned} \int_{\Omega} \left(\frac{\rho}{\beta \Delta t^2} \mathbf{u} \cdot \mathbf{v} + \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \right) + \int_{\partial\Omega_p} \frac{\rho\gamma}{\beta \Delta t} \mathbf{u} \cdot \mathbf{P} \cdot \mathbf{v} = \\ \int_{\Omega} \frac{\rho}{\beta} \left(\frac{1}{\Delta t^2} \mathbf{u}_{\text{old}} \cdot \mathbf{v} + \frac{1}{\Delta t} \dot{\mathbf{u}}_{\text{old}} \cdot \mathbf{v} + \left(\frac{1}{2} - \beta \right) \ddot{\mathbf{u}}_{\text{old}} \cdot \mathbf{v} \right) + \\ \int_{\partial\Omega_p} \left(\frac{\rho\gamma}{\beta \Delta t} \mathbf{u}_{\text{old}} \cdot \mathbf{P} \cdot \mathbf{v} + \left(\frac{\rho\gamma}{\beta} - \rho \right) \dot{\mathbf{u}}_{\text{old}} \cdot \mathbf{P} \cdot \mathbf{v} + \left(\frac{\rho\gamma \Delta t}{2\beta} - \rho \Delta t \right) \ddot{\mathbf{u}}_{\text{old}} \cdot \mathbf{P} \cdot \mathbf{v} \right). \end{aligned}$$

Here, $(\mathbf{u}, \mathbf{v})$ are the FEM trial and test functions, respectively, defined in FEM linear closed space. $(\mathcal{U}, \mathcal{V})$ defined in $[H^1(\Omega)]^3$, for further details see PSD's soildynamics [documentation](#).

Execution Workflow: PSD begins with automated code generation through the PSD_PreProcess utility, which generates problem-specific FEM code based on user specifications.

```
PSD_PreProcess -problem soildynamics -dimension 3 -top2vol-meshing \
-timediscretization newmark_beta -postprocess uav
```

Typical soil properties and time-integration parameters are included:

```
real rho = 1800.0 , // Density
cs = 2300.0 , // S-wave velocity
cp = 4000.0 ; // P-wave velocity

real tmax = 20.0 , // Total time
t = 0.001 , // Initial time
dt = 0.001 ; // Time step
```

The simulation is executed using the parallel solver with the specified number of MPI-processes:

```
PSD_Solve -np 6144 Main.edp
```

Results such as those presented in Figure 1 can be obtained.

Demonstration

Figure 1 presents a regional-scale earthquake simulation of the Cadarache region in France (50 km $\times$ 50 km) performed with PSD (Badri et al., 2024), comprising over one billion degrees of freedom distributed across 6144 MPI-domains on a 540-million-element mesh with 10 m resolution.

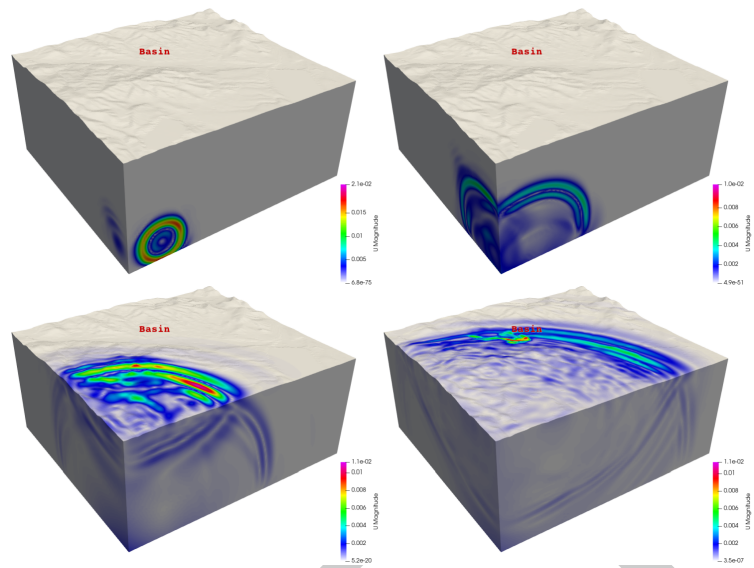


Figure 1: Earthquake simulation of the French Cadarache region showing displacement magnitude at four time steps.

Figure 2 demonstrates fracture mechanics capabilities through quasi-static brittle fracture simulation in a perforated medium (Badri et al., 2021), involving more than 64 million degrees of freedom across 1008 MPI-domains, illustrating detailed damage assessment capabilities.

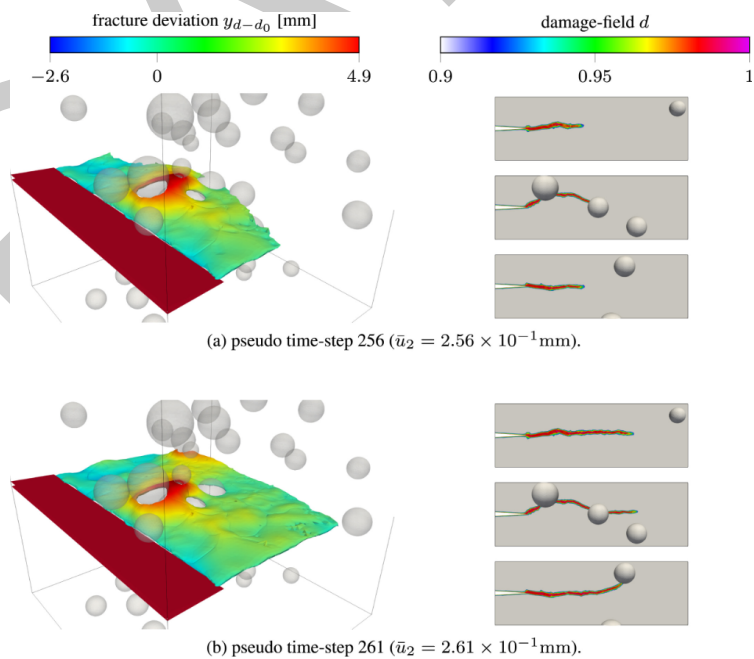


Figure 2: Crack propagation for a perforated medium.

These demonstrations represent significant computational achievements, with problem sizes approaching those required for seismic hazard and risk assessment. The Cadarache region simulation underscores PSD's applicability to real-world earthquake engineering, while the fracture mechanics case illustrates its capability for detailed damage analysis. Further applications, including eikonal non-local gradient damage models (Noigueira et al.,

2023), (Nogueira et al., 2024), highlight PSD's versatility and its potential for comprehensive structural and fracture mechanics research.

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